
Dr Noah Day

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RESEARCH POSITIONS

Postdoctoral Research Fellow | *University of Melbourne*

Sep 2025–Aug 2027

Investigation of a crucial candidate mechanism of Antarctic sea ice loss by predicting rapid ice retreat in response to large Southern Ocean waves

Research Assistant | *University of Adelaide*

March 2025–June 2025

Development of coupled ocean–sea ice–wave configurations for ACCESS-OM3

EDUCATION

Doctor of Philosophy in Mathematical Sciences | *University of Adelaide*

June 2021–July 2025

Modelling the Antarctic Marginal Ice Zone: Classification and Evolution in Response to Storms and Waves

Honours Degree of Bachelor of Mathematical Sciences | *University of Adelaide* Feb 2017–Dec 2020
with First Class Honours

Honours GPA: 6.8/7

Efficient Stochastic Modelling of Vector-Borne Diseases Using Branching Processes

PUBLICATIONS (GOOGLE SCHOLAR CITATIONS: 21)

Day, N. S., Bennetts, L. G., O'Farrell, S. P., Alberello, A., Montiel, F. (2024). Analysis of the Antarctic marginal ice zone based on unsupervised classification of standalone sea ice model data. *Journal of Geophysical Research: Oceans*, 129, e2024JC020953. <https://doi.org/10.1029/2024JC020953>.

Fraser, A., **Day, N. S.**, Wang, Z., Bennetts, L. G., Liu, Q., O'Farrell, S. P., Coleman, R., Voermans, J., Xu, S., Zhu, W., Auger, M., Massom, R., Wongpan, P., Craw, L., Brouwer, J., Toyota, T., Heil, P., Horvat, C. (2026). Revealing the Antarctic marginal ice zone: a decade-long wave-in-ice climatology. *Nature Communications*, 17, 6648, <https://doi.org/10.1038/s41467-026-73203-z>.

Day, N. S., O'Farrell, S. P., Bennetts, L. G. (2026). Model assessment of winter extratropical cyclone short-term impacts on the Antarctic marginal ice zone. EGU sphere [preprint], <https://doi.org/10.5194/egusphere-2025-5209>, *Under revision following peer review at The Cryosphere*.

Constantinou, N., Neme, J., Aguiar, W., Auger, M., Barnes, A. J., Beucher, R., Bhagtani, D., Bull, C. Y. S., Dawson, H., **Day, N. S.**, et al. (n.d.). COSIMA Cookbook: a community-driven resource for ocean and sea-ice modelling science. *Submitted to The Journal of Open Source Education*.

Day, N. S., O'Farrell, S. P., Bennetts, L. G. (n.d.). Emergent wave attenuation regimes from wave-induced breakup of sea ice. *In preparation for submission to Ocean Modelling*.

Day, N. S., Kannadasan, E., O'Farrell, S. P., Fraser, A. D., Meucci, A., Toffoli, A., Bull, C. Y. S., Bennetts, L. G. (n.d.). Quantification of coupled wave–sea ice–ocean interactions on Antarctic sea ice in ACCESS-OM3. *In preparation for submission to JAMES*.

SCIENCE COMMUNICATION

Tracking the Antarctic ice most at risk of breakup and melting. **Day, N. S.**, Bennetts, L. G. 2026. *The Pursuit*.

MIZ-ing in action: how much of Antarctic sea ice is affected by waves? 2026. *Australian Antarctic Program Partnership*.

Presented PhD research at South Australia's largest student STEM expo (Ingenuity) to high school and undergraduate audiences (2022).

GRANTS AND AWARDS

Achievements:

- Andrew Sisson Grant 2026 (\$9,951 AUD)
- University of Melbourne School of Mathematics and Statistics Grant 2026 (\$9,951 AUD)
- Chief Investigator on a National Computational Merit Allocation Scheme (NCMAS) 2026 project, securing 1.7 million core-hours on Australia’s national high-performance computing facilities
- Student oral presentation prize | *IGS International Symposium on Sea Ice (2023)*
- PhD top-up scholarship | Funded by the *Consortium for Ocean-Sea Ice Modelling in Australia* (2021–2024)
- Travel grants:
 - * DR Stranks Travelling Fellowship 2023 (\$3,640 AUD)—Extended research visit to the University of East Anglia, UK
 - * University of Adelaide Graduate Research School Travel Grant 2023 (\$2,500 AUD)—Extended research visit to the University of East Anglia, UK
 - * Quarterly Journal of Mechanics and Applied Mathematics 2023 (£1,300 GBP)—Presentation at the European Geophysical Union General Assembly 2023
 - * Community Ice Code (CICE) Sea Ice Modelling Workshop 2024 (\$2,920 USD funded by the Office of Naval Research)—Los Alamos National Laboratory, USA
 - * neXtSIM Sea Ice Modelling Workshop 2025 (\$2,138 USD)—Brown University, USA
 - * Australian Centre for Excellence in Antarctic Science ECR Travel Grant 2025 (\$1,100 AUD)

Community Contributions and Service:

- Hackathon Prize: Best contribution to the *COSIMA Recipes* Github repository
- International workshops on development of the CICE and neXtSIM sea ice models
- Session convenor at the Australian Antarctic Research Conference 2026
- Reviewer for Journal of Geophysical Research: Oceans, JAMES, and The Cryosphere
- Organiser and Chair of fortnightly meeting series for development of wave modelling in ACCESS-OM3 attendee institutions include University of Melbourne, CSIRO, ACCESS-NRI (2025–ongoing)
- Organised a thesis writing session for postgraduate students in the Melbourne Applied Mathematics and Mathematical Biology Group (2025)

TEACHING

Assistant Lecturer | University of Melbourne

March 2026–July 2026

Lectured for *Linear Statistical Models (MAST30025)*: subject of ~450 students, managed teaching assistants and tutorials, prepared and delivered lectures.

Teaching Assistant | University of Adelaide

Aug 2021–Dec 2024

Practicals, workshops, tutorials, marking and exam invigilation for undergraduate and post-graduate mathematics courses.

- STATS 1000—Mathematics for Data Science I (S2 2024, S1 2023, S2 2022, S2 2021)
- STATS 7022—Data Science Post Graduate (S1 2024)
- MATHS 7027—Mathematical Foundations of Data Science (S1 2024, S2 2023)
- STATS 1000—Statistical Practice I (S1 2024)
- MATHS 1009—Introduction to Financial Mathematics (S1 2023)
- MATHS 1010—Applications of Quantitative Methods in Finance I (S2 2022)
- MATHS 1011—Mathematics IA (S1 2022)

COORDINATION AND SUPERVISION

- Supervision and management of a research assistant for six weeks. The project resulted in completing the preparation of a journal article ready for submission to Geophysical Research Letters
- Offering research support and guidance to several PhD and Masters students

International:

- Climate and Cryosphere (CliC) Open Science Conference (2026)—Wellington, New Zealand
- NeXtSIM: Next generation sea ice model workshop (2025)—Brown University, USA
- CICE Consortium User Workshop (2024)—Santa Fe, USA
- Los Alamos National Laboratory (2024)—Los Alamos, USA
- Ocean Sciences Meeting (2024)—New Orleans, USA
- University of Texas San Antonio Antarctic Sea Ice and Southern Ocean seminar series (2024)—Online
- UK Met Office polar group seminar (2023)—Exeter, UK
- International Symposium on Sea Ice (2023)—Bremerhaven, Germany
- EGU General Assembly (2023)—Vienna, Austria
- University of East Anglia School of Mathematics postgraduate student seminar series (2023)

Domestic:

- Australian Antarctic Research Conference 2026—Hobart, Australia
- Australian Centre for Excellence in Antarctic Science Research Forum 2025—Hobart, Australia
- ACCESS-NRI Workshop 2025—Melbourne, Australia
- Consortium for Ocean-Sea Ice Modelling in Australia Meetings (2022, 2024, 2025)
- Australian Ocean Modelling and Observations Workshop 2024—Canberra, Australia
- University of Adelaide School of Mathematics Postgraduate Student Seminar Series (2022, 2023, 2024)
- Consortium for Ocean-Sea Ice Modelling in Australia Workshop 2022—Hobart, Australia
- Institute for Marine and Antarctic Studies Sea Ice Seminar Series (2022)—Online

REFEREES

Professor Luke Bennetts | *Principal Investigator and PhD Supervisor*

University of Melbourne, luke.bennetts@unimelb.edu.au

Dr Siobhan O'Farrell | *PhD Co-supervisor*

University of Melbourne and CSIRO Aspendale, siobhan.ofarrell@unimelb.edu.au